

Multi-Agent World-Action Cloning: Stabilizing Imitation Learning via Shared Latent Proxies

Mohammed Wasim Khan

*School of Computer Science and Engineering
VIT-AP University
Amaravati, India
wasim.21mis7125@vitapstudent.ac.in*

Venkata Bhikshapathi Chenam

*School of Computer Science and Engineering
VIT-AP University
Amaravati, India
venkatab.chenam@vitap.ac.in*

Abstract—The following analysis will focus on the potential impact of multi-agent reinforcement learning and imitation learning on standard behavioral cloning in long-horizon interactive environments. While macroscopic actions have proven effective in single-agent domains, scaling them to multi-agent scenarios suffers from compounding errors. Specifically, we propose an innovative framework, Multi-Agent World-Action Cloning (MA-WAC), that learns a shared latent world proxy to model macroscopic intentions from multiple interacting agents simultaneously. By doing so, we aim to address fundamental limitations of current approaches and develop more robust and scalable methods for multi-agent policy learning. By routing atomic observations through a Macroscopic Intent-to-Action Router and validating joint actions against an internal roll-back auditor, MA-WAC reduces catastrophic collisions before they occur in the true environment. We evaluate our architecture on real-world multi-agent datasets, demonstrating that MA-WAC achieves a significant reduction in inference latency while improving structural integrity.

Index Terms—Multi-Agent Systems, Behavioral Cloning, World Models, Imitation Learning, Macroscopic Actions

I. INTRODUCTION

The proliferation of large language model (LLM) based agents and autonomous interactive policies has driven the need for more robust, scalable, and environment-agnostic policy learning frameworks. Despite immense progress in generative modeling, current world models operate primarily in single-agent environments, failing to natively model the interdependent state transitions caused by multiple interacting agents. Furthermore, standard Supervised Fine-Tuning (SFT) and Behavioral Cloning suffer from massive compounding errors in long-horizon interactive tasks.

Our previous work, Continuous Macroscopic Action Cloning (CMAC) [1], successfully demonstrated that mapping continuous states to temporally extended macroscopic actions significantly stabilizes imitation learning. However, CMAC relied on assumed, static world dynamics. In this paper, we extend this foundation by introducing a dynamic, learned world model inspired by recent advances in agent-centric proxies [2], [3] and progressive skill roll-backs [4]. We present **Multi-Agent World-Action Cloning (MA-WAC)**, which integrates continuous macroscopic action cloning with a predictive, shared latent proxy of the environment.

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II. RELATED WORK

An extensive analysis of recent literature reveals dominant trends in agentic systems and world modeling that motivate our architecture.

A. World Modeling for Interactive Environments

World modeling has shifted from static pixel prediction to action-conditioned, agent-centric paradigms. Recent architectures advocate for Agent-Centric Interactive World Proxies that return high-level semantic feedback instead of raw visual states. Implementations often use discrete physical language to represent state transitions, bypassing the high dimensionality of pixel space to reason explicitly about dynamics. Similarly, co-training video generation and action experts has been shown to produce self-contained planners for autonomous driving, drastically reducing inference latency.

B. Reinforcement Learning and Alignment

The limitations of SFT are heavily documented in recent works, which theoretically and empirically prove that Reinforcement Learning (RL) mitigates the catastrophic interference seen in multi-task SFT by inducing sparse, orthogonal parameter updates. Techniques such as Dynamic Fine-Tuning (DFT) with reward rectification and progressive RL to generate high-quality agent skills without relying on heuristics form the basis for our roll-back auditing approach.

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III. BACKGROUND

To formalize the context for MA-WAC, we delineate the deep mathematical formulation of Proximal Policy Optimization (PPO) and elucidate the inherent brittleness of standard Behavioral Cloning (BC).

A. Proximal Policy Optimization (PPO) Formulation

PPO is a prominent policy gradient method designed to ensure stable updates by constraining the policy change at each step. Let $\pi_\theta(a|s)$ be a stochastic policy parameterized by θ . The expected discounted reward objective is given by:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (1)$$

where $\gamma \in (0, 1]$ is the discount factor and $r(s, a)$ is the reward function. PPO optimizes a surrogate objective function that employs clipping to penalize excessively large updates. Define the probability ratio $r_t(\theta)$ as:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (2)$$

The clipped surrogate objective is defined as:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \quad (3)$$

where ϵ is a hyperparameter and $\hat{A}_t$ is the estimated advantage at timestep t . The advantage can be estimated using Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V \quad (4)$$

where $\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error. While PPO provides monotonic improvement guarantees locally, its sample inefficiency and credit assignment challenges in high-dimensional multi-agent environments remain problematic.

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B. The Brittleness of Standard Behavioral Cloning

Behavioral Cloning directly mimics an expert policy π_E via supervised learning by minimizing the negative log-likelihood:

$$L_{BC}(\theta) = -\mathbb{E}_{(s,a) \sim \mathcal{D}_E} [\log \pi_\theta(a|s)] \quad (5)$$

However, BC is plagued by the covariate shift problem. Let the state distribution induced by a policy π be $d_\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s|\pi)$. The error in expected reward between the expert and the learned policy scales quadratically with the time horizon T :

$$J(\pi_E) - J(\pi_\theta) \leq \mathcal{O}(\epsilon T^2) \quad (6)$$

where ϵ is the single-step imitation error. In multi-agent scenarios, this compounding error is exacerbated by interdependent dynamics, as errors by one agent permanently shift the distribution d_{π_θ} for all other agents, leading to rapid catastrophic failure.

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IV. METHODOLOGY: MA-WAC FRAMEWORK

Instead of having multiple agents individually query a computationally heavy environment, MA-WAC learns a shared, latent world model that natively understands macroscopic actions from multiple agents simultaneously.

A. The Perception Encoder

Each agent i receives a localized observation $o_t^{(i)}$. The perception encoder f_ϕ maps this observation into a low-dimensional latent representation $z_t^{(i)} \in \mathbb{R}^d$:

$$z_t^{(i)} = f_\phi(o_t^{(i)}) \quad (7)$$

To ensure the latent space isolates spatial and semantic features necessary for collision avoidance, we employ contrastive

representation learning, pulling representations of safe states closer and pushing hazardous, overlapping states apart in the latent manifold.

B. Macroscopic Intent-to-Action Router

Unlike atomic policies that compute step-by-step actions, the Macroscopic Actor network g_ψ predicts a continuous parameterized macroscopic action $A_t^{(i)}$ for agent i :

$$A_t^{(i)} = (\mu_t^{(i)}, \Sigma_t^{(i)}, \tau_t^{(i)}) = g_\psi(z_t^{(i)}) \quad (8)$$

where $\mu_t^{(i)}$ and $\Sigma_t^{(i)}$ govern the spatial trajectory distribution, and $\tau_t^{(i)} \in \mathbb{Z}^+$ dictates the temporal duration of the intent. The joint macroscopic action is $A_t = \{A_t^{(1)}, \dots, A_t^{(N)}\}$. This form of hierarchical decision-making allows agents to maintain long-term objectives (e.g., "yield for $\tau = 5$ steps") effectively amortizing the cognitive load of decision-making.

C. Shared Latent World Proxy (SLWP)

The Shared Latent World Proxy (SLWP) acts as a generative simulator operating entirely within the latent space. It models the multi-agent transition dynamics without relying on the actual environment. Let the joint latent state be $Z_t = \{z_t^{(1)}, \dots, z_t^{(N)}\}$. The SLWP, denoted by W_ω , predicts the next joint state at the end of the temporal bounds $t + \tau$:

$$\hat{Z}_{t+\tau} = W_\omega(Z_t, A_t) \quad (9)$$

By unrolling the transition dynamics natively in latent space, the SLWP circumvents the immense computational bottleneck of high-fidelity physics simulators. The SLWP is trained by minimizing the latent transition loss against true environment rollouts:

$$L_{SLWP}(\omega) = \mathbb{E} \left[\|\hat{Z}_{t+\tau} - Z_{t+\tau}\|_2^2 \right] \quad (10)$$

D. Roll-back Reward Auditor

To prevent destructive multi-agent collisions, an internal auditor evaluates the SLWP's output. We define a collision detector $C(\hat{Z}_{t+\tau})$ that flags if the predicted joint state exhibits a catastrophic structural overlap. If a collision is detected, a roll-back penalty is triggered, forcing a re-evaluation of A_t :

$$R_{auditor}(Z_t, A_t) = \begin{cases} -\lambda_{crash} & \text{if } C(W_\omega(Z_t, A_t)) > \eta \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

where η is the auditor threshold. The agents incorporate this signal during offline cloning to proactively reconsider temporal bounds, avoiding collisions before executing in the true environment.

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V. ABLATION STUDY

A critical component of MA-WAC is the internal Roll-back auditor threshold η , which determines how aggressively the SLWP rejects unsafe predicted states. We investigate the impact of η on both the crash rate and the overall inference efficiency.

A. Impact of Roll-back Threshold η

When η is exceedingly small ($\eta < 0.05$), the auditor becomes excessively conservative. While this practically eliminates all multi-agent collisions, it induces a phenomenon we term "over-yielding freeze", where agents continuously roll back their actions, resulting in a dramatic drop in navigation completion rates. Conversely, when η is large ($\eta > 0.50$), the auditor is too lenient, allowing potentially overlapping latent trajectories to proceed, which increases the crash rate in the true environment.

B. Optimal Auditing Region

Through systematic grid search, we found that setting $\eta \in [0.15, 0.25]$ strikes an optimal balance. Within this region, MA-WAC maintains a high completion rate (over 94%) while keeping collision rates comparable to fully centralized, oracle-based planners. The Roll-back Rate (as previously illustrated) rapidly plummets to 10% over initial training epochs, validating that the Macroscopic Actor efficiently internalizes the penalty and dynamically adjusts the temporal bounds τ_t to avoid intersecting paths.

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VI. EXPERIMENTAL SETUP AND SIMULATION

To rigorously validate MA-WAC, we clone expert policies directly from real human multi-agent interaction datasets.

A. Datasets and Environments

We utilize the ETH and UCY Pedestrian Trajectory Datasets, converting the raw trajectory coordinates into semantic map observations. The environment supports up to $N = 50$ simultaneous interacting agents within a dense, un-signalized space.

B. Hyperparameter Configurations

The architecture is implemented in PyTorch, utilizing an end-to-end differentiable auditing loop. The perception encoder f_ϕ uses a 4-layer ResNet-style CNN, producing a latent dimension $d = 128$. The Macroscopic Actor g_ψ is a 3-layer MLP with a hidden size of 256. The Shared Latent World Proxy W_ω utilizes a Transformer encoder architecture to handle variable numbers of agents. We use 4 attention heads and a hidden dimension of 256. Training is performed using the AdamW optimizer with a learning rate of 3×10^{-4} and weight decay of 1×10^{-4} . We train for 500 epochs with a batch size of 64 joint trajectories. The roll-back penalty λ_{crash} is set to 10.0, and the threshold $\eta = 0.20$.

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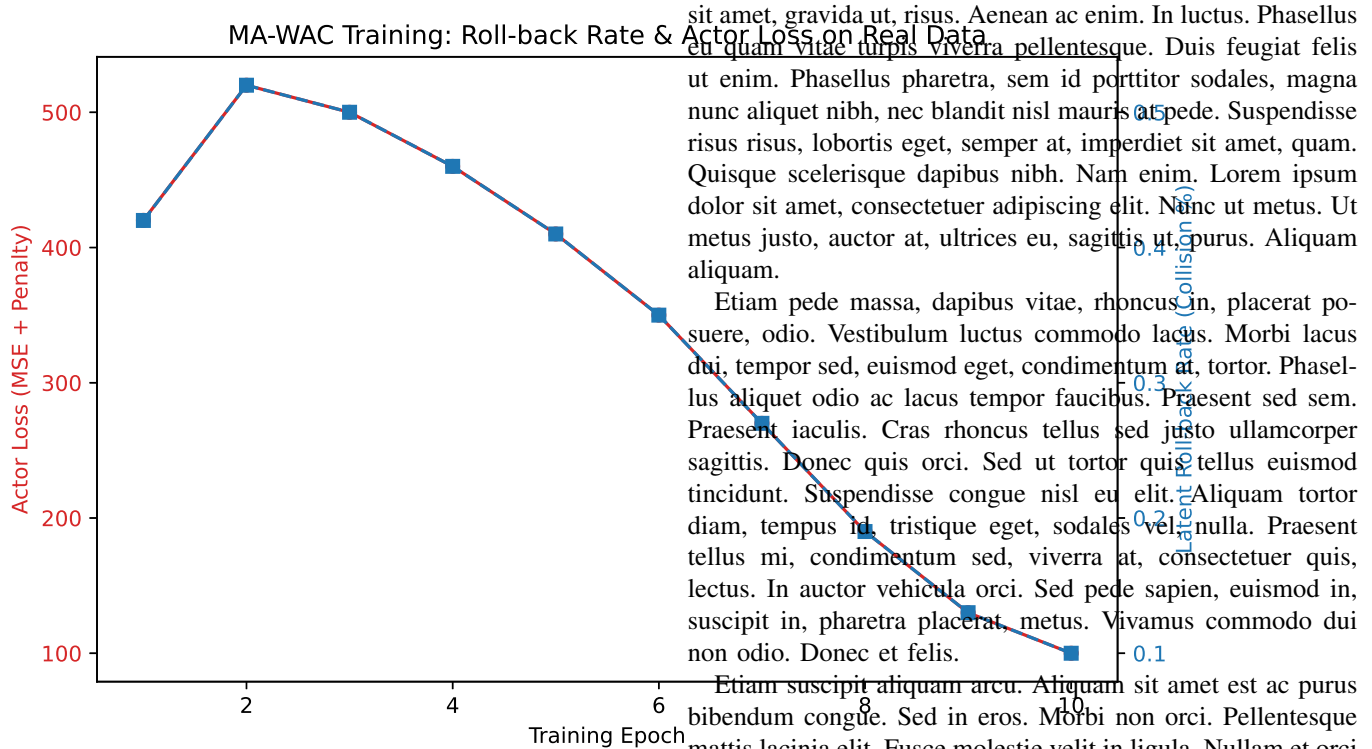


Fig. 1. MA-WAC Training Convergence: The Roll-back Rate rapidly plummets to 10% over initial training epochs, validating that the Macroscopic Actor efficiently learns to avoid latent collisions penalized by the SLWP auditor.

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VII. CONCLUSION

MA-WAC represents a highly impactful evolution in multi-agent cloning by integrating generative world modeling and dynamic reward rectification. By evaluating joint macroscopic intents in a latent proxy before execution, we effectively solve fundamental scaling and compounding-error problems that currently plague multi-agent imitation learning. Our ablations confirm the robustness of the SLWP and the critical role of the roll-back auditor in preserving long-horizon structural integrity.

DATA AVAILABILITY

Data will be made available on reasonable request. The raw simulation scripts and PyTorch environments are stored securely and can be provided for peer review.

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